

News Article Clustering using Word2Vec

Problem Statement

You are given a collection of short news articles (or headlines). The task is to **group similar articles together** using **unsupervised learning**. Instead of directly clustering words, we will:

1. Represent each document as the **average of its Word2Vec embeddings**.
2. Apply **clustering (KMeans)** on the document embeddings.
3. Evaluate results qualitatively by printing sample clusters of articles.

This demonstrates:

- Word2Vec embeddings at the **document level**.
 - How embeddings capture **semantic similarity**.
 - Using **unsupervised ML with NLP**.
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✅ Step-by-Step Guide

1. **Import libraries**
2. **Load dataset** (a small news headline dataset – can use AGNews, or a CSV with short texts)
3. **Preprocess text** (tokenize, lowercasing, stopword removal)
4. **Train Word2Vec embeddings** using gensim
5. **Create document embeddings** (average of word embeddings)
6. **Apply clustering** using KMeans
7. **Visualize clusters** with t-SNE or PCA
8. **Evaluate** by printing some headlines from each cluster

Expected Outcome

- The articles will be grouped roughly into categories like **Technology**, **Finance**, and **Sports/Space**.
- Example clusters:
 -  *Apple unveils new iPhone, Samsung releases foldable phone*
 -  *Stock markets drop, Investors worried*
 -  *NASA mission, SpaceX launch*

This shows how **Word2Vec embeddings** + **clustering** can capture **semantic meaning** beyond just keywords.